SRS V1 — Scaffolded Student Template

Complete incrementally from Week 8 backlog and Week 9 elicitation. Do not invent facts; mark uncertainties as TBD.

Team / Project: _____ Version: 0.1 Date: _____ Reviewers: _____

How to use this scaffold

Start with needs and stories, then progressively specify functional requirements, use cases, and tests. These are linked artifacts, not a mandatory one-way sequence. In Week 9, aim for a credible V1 rather than a complete final SRS.

Stage	Minimum evidence	Done?
A — Discover	Stakeholders, needs, scope, assumptions	[ ]
B — Elaborate	3–5 stories, 3–5 FRs, 2 use cases, 2 NFRs	[ ]
C — Validate	Acceptance criteria, trace links, peer-review fixes	[ ]

1. Introduction, purpose, and scope P

Instructor/student prompt: Describe problem, intended users, business value, in scope / out of scope.

Fill in

Problem: [Parcel delivery operations can face delays, inaccurate tracking information, and difficulties managing orders, payments, and deliveries.]
System goal: [To develop a parcel management system that improves delivery efficiency, tracking accuracy, and communication between customers and staff.]
In scope:

- Customer registration and login
- Parcel booking and order management
- Shipping fee calculation
- Parcel tracking and status updates
- Pickup and delivery management
- Payment and COD management
- Employee role and access management
- Customer support

Out of scope:

- International Shipping – No international parcel delivery.
- Online Marketplace and Product Sales – No online product buying or selling.
- Automated Route Optimization – No automatic calculation of the best delivery routes.

2. Stakeholders and needs P

Instructor/student prompt: Use evidence from interviews, Week 8 vision or clearly labeled assumptions.

Need ID	Stakeholder	Need / pain point	Evidence / TBD
N-01	Customer (Sender)	Needs to create orders and track parcels easily	interview
N-02	Recipient	Needs to check delivery status and receive parcels on time	interview
N-03	Counter Staff	Needs to register parcels and process payments accurately	interview
N-04	Hub staff	Needs to sort parcels and update delivery statuses efficiently	interview
N-05	Finance Staff	Needs accurate payment records and COD tracking	interview
N-06	Manager	Needs to monitor delivery performance and service quality	interview
N-07	Customer Service Staff	Needs quick access to shipment information to resolve inquiries	interview
N-08	System Administrator	Needs to manage user permissions and prevent unauthorized access	interview

3. Context, assumptions and constraints D

Instructor/student prompt: Identify system boundary, key policies, dependencies, and unresolved questions.

ID	Assumption / constraint	Owner	How to validate

A-01	Tracking number creation time is TBD.	Counter Staff	Ask when tracking numbers are created.
A-02	Pickup appointment changes are TBD.	Pickup Rider / Manager	Ask whether customers can change their pickup time.
A-03	Parcel status update permissions are TBD.	Manager / System Admin	Confirm which roles can update each status.

4. User stories and priorities D

Instructor/student prompt: Write stories for value and conversation, not as substitutes for all system requirements.

Story ID	As a / I want / so that	Priority	Need ID
US-01	As a customer, I want to track my parcel, so that I know its delivery status.	Must	N-01
US-07	As a finance staff member, I want to view COD records, so that I can transfer money to the correct sender.	Must	N-05

5. Functional requirements A

Instructor/student prompt: Turn agreed behavior into unambiguous, testable shall statements; cite related stories where relevant.

FR ID	The system shall...	Priority	Story / source
FR-01	The system shall allow senders and receivers to track parcel delivery status using a valid tracking number.	Must	US-01
FR-02	The system shall allow pickup riders to view scheduled pickup appointments.	Must	US-02
FR-03	The system shall allow riders to access customer contact information required for pickup and delivery operations.	Should	US-03
FR-04	The system shall allow authorized employees to update parcel status throughout the delivery process.	Must	US-04

FR-05	The system shall allow customers and counter staff to create delivery orders by providing sender, receiver, parcel, and service information.	Must	US-05
FR-06	The system shall calculate shipping fees based on parcel weight, dimensions, destination zone, and service type.	Must	US-06
FR-07	The system shall require payment confirmation before an order is successfully created and shall record payment and COD transaction information for each order.	Must	US-07
FR-08	The system shall provide managers with access to delivery records and operational reports.	Should	US-08
FR-09	The system shall allow customer service staff to search and view shipment information in order to respond to customer inquiries.	Must	US-09
FR-10	The system shall allow system administrators to manage user accounts, roles, and permissions.	Must	US-10
FR-11	FR-11 The system shall automatically generate a unique tracking number when a delivery order is successfully created.	Must	Order Creation / Tracking Process

6. Use cases (two) B

Instructor/student prompt: Choose meaningful actor-goal interactions; one use case may cover several FRs/stories.

UC-01: [Create Shipping Order]

Primary actor: [Sender (Customer)] Trigger: [Sender requests to create a shipping order.]
Preconditions: [Sender can access the order creation page.]
Main success flow: 1. [Sender enters sender and receiver information.] 2. [Sender enters parcel details, including weight and dimensions.] 3. [Sender selects a delivery service.] 4. [The system validates the information and calculates the shipping fee.] 5. [Sender confirms the shipping order.] 6. [System creates the order, generates a unique tracking number, and displays the order details.]
Alternate : A1: Required information is missing or invalid → System rejects the submission and requests correction.
Alternate : A2: Sender cancels before confirmation → No shipping order is created.
Postconditions: [A new shipping order and parcel record are created with a unique tracking number, or no order is created if the process fails or is cancelled.]
Related FRs: [FR-05, FR-06, FR-07, FR-11]

UC-02: [Track Parcel Status]

Primary actor: [Sender (Customer)] Trigger: [Sender requests to check a parcel's delivery status.]
Preconditions: [A parcel record exists, and the sender is authorized to access its tracking information.]
Main success flow: 1. [Sender opens the parcel tracking page.] 2. [Sender selects an existing parcel order.] 3. [System retrieves the parcel's latest tracking information.] 4. [System displays the current status and available tracking history.]
Alternate : A1: Parcel record cannot be found → System displays an error message.
Alternate : A2: No tracking updates are available → System displays the existing parcel status without additional tracking history.
Postconditions: [Sender views the available parcel tracking information. No parcel information is modified.]
Related FRs: [FR-01]

7. Non-functional requirements A

Instructor/student prompt: Specify measurable qualities or constraints; state how to verify them.

NFR ID	Requirement + threshold	Verification method	Source / assumption
NFR-01	The system shall respond to parcel tracking and shipment search requests within **2 seconds for at least 95% of requests** under normal operating conditions.	Performance Test	US-01, US-09
NFR-02	The system shall support at least **100 concurrent users** without significant	Load Test	Assumption

	performance degradation.		
NFR-03	The system shall maintain **99% availability** during business hours.	Availability Monitoring	Assumption
NFR-04	The system shall require user authentication before allowing access to protected functions and sensitive information.	Security Test	US-03, US-07, US-10
NFR-05	The system shall enforce **Role-Based Access Control (RBAC)** for Customer, Rider, Customer Service Staff, Finance Staff, Manager, and System Administrator roles.	Access Control Test	US-08, US-09, US-10
NFR-06	The system shall protect confidential customer information and financial records from unauthorized access.	Security Audit	Privacy Policy
NFR-07	The system shall record audit logs for parcel status updates, payments, and account management activities.	Audit Log Review	US-04, US-07, US-10
NFR-08	The system shall perform daily database backups and support data recovery in case of system failure.	Backup & Recovery Test	Business Continuity Requirement

8. Acceptance criteria and tests D

Instructor/student prompt: Include happy path and negative path; acceptance criteria can derive from stories, FRs, and use cases.

AC / Test ID	Given / When / Then or test steps	Expected result	Linked IDs

AC-01	Given a parcel exists, when a user enters its valid tracking number, then the system shows its delivery status.	The correct parcel status appears.	US-01 / FR-01
AC-02	Give a tracking number does not exist, when a user searches for it, then the system shows a “Parcel not found” message.	No parcel details appear.	US-01 / FR-01
TC-01	Prepare a parcel record. Enter its tracking number and check the result.	The displayed status matches the parcel record.	AC-01
TC-02	Enter a tracking number that does not exist and check the result.	“Parcel not found” appears.	AC-02

9. Traceability and review B

Instructor/student prompt: Every high-priority requirement needs an evidence source and planned verification.

Need	Story (optional)	FR	Use case (optional)	AC / test
N-01	US-01	FR-01	UC-02	AC-01 / TC-01
N-02	US-02	FR-02	-	AC-02 / TC-02
N-03	US-03	FR-03	-	AC-03 / TC-03
N-04	US-04	FR-04	-	AC-04 / TC-04
N-05	US-05	FR-05	UC-01	AC-05 / TC-05
N-06	US-06	FR-06	UC-01	AC-06 / TC-06
N-07	US-07	FR-07	UC-01	AC-07 / TC-07
N-08	US-08	FR-08	-	AC-08 / TC-08
N-09	US-09	FR-09	-	AC-09 / TC-09
N-010	US-10	FR-10	-	AC-10 / TC-10

Peer-review log

Reviewer: [ ] Date: [ ]
Finding: [ambiguous / missing / inconsistent / untestable] Fix / owner: [ ]
Open questions to resolve in Week 10: [ ]